Carter Gladden

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Education

Loyola Marymount University — Los Angeles, CA
Bachelor of Science in Computer Science, Minor in Mathematics - GPA: 4.00 — *Aug. 2025 – May 2028*

University of Oregon — Eugene, OR
Bachelor of Science in Computer Science (transferred) - GPA: 3.95 — *Sept. 2024 – Jun. 2025*

Professional Experience

Data Engineering Intern — Jun. 2025 – Aug. 2025
Kollective Technology — *Bend, OR*

- Integrated Marimo Python notebooks into the data engineering workflow, designing and testing templates that connect to Trino Iceberg and Citus PostgreSQL data lakes
- Built querying, transformation, and visualization functionality using Pandas, Ibis, Altair, and advanced JSON parsing to deliver analytics insights
- Drove adoption across a team of 10 engineers through demos and presentations, making analytics code more reactive, reproducible, and git-friendly

Lab Teaching Assistant — Aug. 2025 – Present
Loyola Marymount University — *Los Angeles, CA*

- Guide students through debugging and CS coursework, adapting explanations to individual skill levels while preserving independent problem solving

Research Experience

Simulating Dashboard Interactions — May 2026 – Present
University of Washington — *Seattle, WA and Remote*

- Developed a database benchmark that simulates interactive data exploration for more realistic query workloads
- Designed a text-based dashboard representation enabling LLM agents to interpret and act on visualizations
- Built an LLM agent environment and harness system to interact with visualization dashboards
- Co-authoring a research paper documenting the framework and its evaluation

GRNsight — Aug. 2026 – Present
Loyola Marymount University — *Los Angeles, CA*

- Contributing to GRNsight, an open-source Node.js/Express and PostgreSQL web application for visualizing gene regulatory networks, maintained across a production client and a React/Vite rewrite

Projects

LA AQI Forecasting | *Python, scikit-learn, FastAPI, Pandas, Docker* — GitHub

- Built an ML pipeline forecasting Los Angeles air quality from real-time PurpleAir sensor data, training a Random Forest regressor evaluated on MAE, RMSE, and R^2

Habit Tracker | *Next.js, TypeScript, Firebase, Tailwind CSS* — Live | GitHub

- Built and deployed a mobile-first PWA habit tracker with streak tracking, per-habit journaling, and push notifications
- Implemented Google and email auth, real-time Firestore sync with security rules, and server-side token verification via Firebase Admin

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL (PostgreSQL), C/C++, Assembly, HTML/CSS
Frameworks: React, Next.js, Node.js/Express, FastAPI, Streamlit, Tailwind CSS
Libraries: pandas, NumPy, scikit-learn, PyTorch, Pydantic, Ibis, Altair, pytest
Developer Tools: Git, Docker, GitHub Actions, Firebase/Firestore, Vercel, Trino, LaTeX

Activities and Interests

Activities: ACM, Creare Service Organization (LMU) — **Interests**: rock climbing, surfing, golf